

CWA SCIENCE CLASSES

Concept With Abhishek | Class 11 — Physics

Chapter 2: Motion in a Straight Line

1. Kinematics deals with the description of motion without considering the cause of motion.
2. Position, Path length & Displacement: Displacement can be zero, positive or negative; path length is always positive.
3. Average velocity = Total displacement / Total time. Average speed = Total path length / Total time.
4. Instantaneous velocity = derivative of position w.r.t time (dx/dt).
5. Acceleration = dv/dt . Uniform acceleration equations: $v = u + at$, $x = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2ax$.
6. Relative velocity: velocity of one object as observed from another moving object; $v_{AB} = v_A - v_B$.
7. Graphs: slope of x-t graph gives velocity; slope of v-t graph gives acceleration; area under v-t graph gives displacement.